BUILDING THE TWIN

Static inputs

Module 3 · Capturing the shape of the network

Transport Twin Study Path

Ways to capture reality

Laser scanning

Terrestrial: millimetre detail

Mobile: whole corridors

Airborne: wide areas

Photogrammetry

Drone images to 3D

Cheap and fast

Needs good light and control points

From points to a model



Network models need more than geometry

- Nodes, links, lanes and allowed turns
- Signals and their programmes
- Zones and origin–destination demand

BIM, IFC 4.3 and CityGML

- IFC 4.3: roads, rail, bridges (ISO 16739-1:2024)
- CityGML: city objects at levels of detail
- OpenDRIVE: lane-level roads for driving simulators

Rule of thumb

Capture once, use many times

Plan coordinate systems, accuracy and formats before the survey